



Research Paper

Balance sheet decomposition of crypto, stablecoin, central bank digital currency and tokenized assets

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ABSTRACT

Cryptocurrencies, stablecoins, central bank digital currencies and tokenized real-world assets are reshaping the balance sheets of central banks, commercial banks, nonbank financial intermediaries and households, yet the existing literature examines them in isolation. This paper constructs a unified, multi-entity balance sheet framework that incorporates all four asset classes via a six-stage stylized process and gives background showing the evolution of each stage. It makes a threefold contribution to the literature. First, we reveal cross-asset transmission channels, such as stablecoin runs propagating through money market funds into sovereign debt markets, that are invisible in single-asset studies. Second, we show that permissioned and permissionless blockchain issuances produce fundamentally different balance sheet dynamics: neutral liability substitution in the former versus unbounded off-balance-sheet leverage, analogous to pre-2008 shadow banking vehicles, in the latter.

Third, we document a decentralized finance credit multiplier through which bank deposits, once tokenized and reused on-chain, generate multiple layers of credit beyond prudential regulation.

Keywords: cryptocurrency; stablecoin; central bank digital currency; balance sheet; tokenized assets; banking.

1 INTRODUCTION

Driven by the proliferation of blockchain technology and distributed ledger systems, the global financial architecture is undergoing a paradigm shift, moving beyond the traditional fiat-centric order. As of January 28, 2026, the total market capitalization of cryptocurrencies reached US\$3 trillion, with Bitcoin firmly establishing itself as a core asset class within institutional portfolios.

The regulatory landscape witnessed a watershed moment in 2025. The enactment of the Guiding and Establishing National Innovation for US Stablecoins Act (Genius Act) in the United States, the Markets in Crypto-Assets (MiCA) Regulation in the European Union and the Stablecoins Ordinance in Hong Kong provided long-awaited legal clarity for issuers. Consequently, the stablecoin sector experienced explosive growth: by January 2026, its market capitalization had surged to US\$296.86 billion, supported by a user base of 227.16 million holders.¹

Meanwhile, the trajectory of central bank digital currencies (CBDCs) has shown significant geopolitical divergence. While the United States imposed a ban on CBDC issuance under President Trump in January 2025, adoption across the rest of the world has accelerated. At the time of writing, three jurisdictions had officially launched their own CBDCs, and 49 countries were conducting active pilot programs. Further, the tokenization of real-world assets (RWA), which allows on-chain investors to manage traditional assets, has grown to US\$23 billion. On January 28, 2026, the US Securities and Exchange Commission (SEC) released a staff statement on tokenized securities clarifying how federal securities laws apply to on-chain asset representations, while on March 17, 2026, the SEC and the Commodity Futures Trading Commission jointly issued guidance that established a digital asset taxonomy, helping to clarify the distinct regulatory scopes and jurisdictions of both agencies in regulating tokenized assets and addressing their joint roles (US Securities and Exchange Commission 2026a,b).

¹ URL: <https://defillama.com/stablecoins>.

These emerging digital assets (cryptoassets, stablecoins, CBDCs and RWA tokens) are not merely new investment vehicles; they fundamentally restructure the balance sheets of central banks, financial intermediaries and households. This paper employs a balance sheet analysis to rigorously examine their structural impacts.

This paper makes three contributions to the literature.

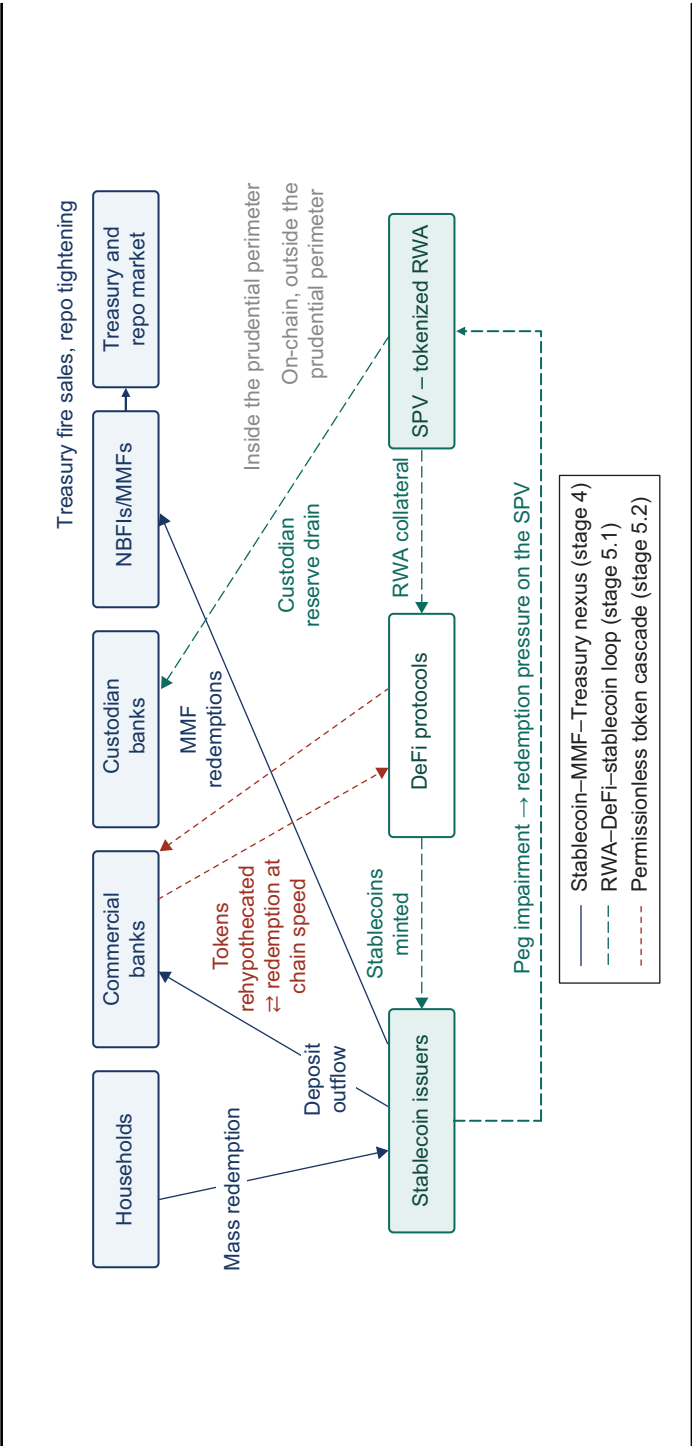
- (1) To the best of our knowledge, this is the first study to analyze all four digital asset classes within a single, internally consistent balance sheet framework. By tracking assets and liabilities across multiple entities simultaneously, we reveal transmission channels that are invisible when each asset is studied in isolation, by showing, for example, how a stablecoin run can propagate through money market funds (MMFs) into the Treasury market, or how a crypto crash can cascade through decentralized finance (DeFi) into bank-issued tokenized assets.
- (2) We identify a novel distinction between permissioned and permissionless blockchain issuances that determines whether bank-issued digital tokens reinforce the existing financial architecture or catalyze a new generation of shadow banking. We show that the same instrument (a tokenized deposit) produces fundamentally different balance sheet dynamics depending on the underlying infrastructure: neutral liability substitution on a permissioned chain versus regulatory reclassification and unbounded off-balance-sheet leverage on a permissionless chain.
- (3) We document a DeFi credit multiplier channel through which a single dollar of bank deposits, once converted into an RWA token and reused through decentralized protocols, can support multiple layers of on-chain credit that exist entirely outside the perimeter of prudential regulation. Recent work has modeled the fragility of layered protocols from within the protocol economy (see, for example, Lee 2026); what the balance sheet framework adds is the funding origin of those layers and the regulated institutions on which they ultimately fall.

Figure 1 maps the three channels that organize the analysis; each is developed at the stage indicated in the legend.

2 LITERATURE REVIEW

The academic and regulatory literature has largely examined the four digital asset classes in distinct, often isolated, streams of research. In this section we summarize this literature.

FIGURE 1 Cross-asset transmission channels between the regulated banking system and the on-chain ecosystem.



2.1 Cryptocurrencies: from speculative assets to reserve drains

Early research on cryptocurrencies primarily focused on their nature as a medium of exchange or as a speculative asset class. Yermack (2015) provides an influential assessment, arguing that Bitcoin fails to satisfy the three classical functions of money due to its extreme volatility. This view is corroborated by Baur *et al* (2018), who demonstrate that Bitcoin is predominantly used as a speculative investment rather than a means of payment. However, as institutional adoption has grown, the focus has shifted toward the systemic footprint in the banking sector. While Liu and Tsyvinski (2021) established that crypto returns are driven by network-specific momentum rather than macro factors, Sockin and Xiong (2023) analyzed how decentralized platforms create channels for price shocks to propagate into traditional markets. A critical gap addressed in our paper is the “digital drain” effect: as households reallocate wealth from bank deposits to cryptoassets, they trigger a structural contraction in commercial bank reserves (a phenomenon we formalize in stages 2 and 3 of our balance sheet analysis).

2.2 Stablecoins: deposit aggregators and shadow banking

The stablecoin literature has coalesced around the view that these instruments are a modern incarnation of private money creation, with structural parallels to historical “wildcat banking” (Gorton and Zhang 2023; Luck 2025). The 2023 USD Coin (USDC) depegging serves as a case study of how stablecoin reserves create a direct transmission channel between the crypto ecosystem and the traditional banking system (Gorton and Zhang 2024).

From a systemic perspective, Liao and Caramichael (2022) provide the most relevant framework by defining stablecoin issuers as deposit aggregators. These entities collect retail liquidity and channel it into MMFs and short-term Treasuries rather than returning it to the banking system as deposits. This mechanism, which we detail in stage 4, accelerates the migration of credit intermediation from commercial banks to nonbank financial intermediaries (NBFIs), effectively narrowing the traditional banking footprint.

2.3 CBDCs and disintermediation

The central debate regarding CBDCs revolves around the risk of bank disintermediation. Brunnermeier and Niepelt (2019) establish a foundational equivalence result, suggesting that a CBDC need not alter credit allocation if the central bank “passes through” the acquired deposits. In contrast, Fernández-Villaverde *et al* (2021) and Kumhof and Noone (2021) argue that a retail CBDC could erode the deposit base and raise funding costs, particularly for small banks (Whited *et al* 2022).

More recent work by Bidder *et al* (2024) distinguishes between “slow disintermediation” and “fast disintermediation”. This paper extends this discourse by incorporating the new development of CBDC (ie, e-CNY 2.0) as tokenized deposits. As noted by Bouis *et al* (2024), such designs may mitigate disintermediation by keeping the liability on the commercial bank’s balance sheet while utilizing a distributed ledger technology for settlement efficiency. We extend the framework of Armas and Singh (2022) to encompass the new development of CBDCs.

2.4 Tokenized RWA: the frontier of collateral transformation

The tokenization of RWA represents the newest frontier in blockchain integration. While Catalini and Gans (2020) analyze the reduction in verification costs, empirical studies such as that by Kreppmeier *et al* (2023) highlight persistent liquidity challenges in tokenized real estate.

A critical gap in the existing RWA literature is the absence of systematic balance sheet analysis. Most research (see, for example, Mirdala 2025) focuses on market microstructure or legal frameworks. Our analysis in stage 5.1 fills this gap by distinguishing between financial RWA and nonfinancial RWA. We argue that the “on-chaining” of traditional assets such as Treasuries changes the velocity of collateral and further pressures the traditional bank deposit model.

2.5 Identifying and filling the gap in the literature

The literature reviewed above reveals a significant structural gap. Each of the four asset classes (cryptocurrencies, stablecoins, CBDCs and RWA) has been studied within its own disciplinary silo: the crypto literature adopts an asset pricing lens without examining the deposit-drain effect on commercial banks; the stablecoin literature analyzes issuer reserve composition without tracing the complete funding chain from household deposits through MMFs back to the banking system; the CBDC literature models disintermediation risk without considering the simultaneous competitive pressure from stablecoins and tokenized deposits; and the RWA literature focuses on market design and legal frameworks without examining the balance sheet consequences for the entities that ultimately provide the underlying liquidity.

More critically, to the best of our knowledge, no existing study has examined the cross-asset transmission channels that connect these instruments through the banking system’s balance sheet. A stablecoin run does not merely affect the stablecoin issuer; it triggers MMF redemptions, which pressure the short-term Treasury market, which in turn affects the collateral base of tokenized RWA products. A crypto crash does not merely reduce household wealth but cascades through DeFi protocols into bank-issued tokenized assets, triggering liquidations that propagate back to the

originating bank's reserve position. These second-order and cross-asset effects are invisible when each instrument is analyzed in isolation.

This paper addresses this lack along three dimensions. First, we construct a novel unified, multi-entity balance sheet framework that simultaneously incorporates all four asset classes, enabling a comparative analysis of their differential impacts on the banking system structure. Second, we identify a previously unexamined distinction (the choice between permissioned and permissionless blockchain infrastructure) that fundamentally determines whether bank-issued digital liabilities reinforce or undermine the existing regulatory architecture, drawing an explicit structural parallel with the off-balance-sheet vehicles that amplified the 2007–9 global financial crisis. Third, we document a DeFi credit multiplier channel linking RWA tokenization to stablecoin minting, through which a single dollar of bank deposits can generate multiple layers of on-chain credit beyond the reach of prudential regulation: a mechanism that has no precedent in the traditional shadow banking literature.

3 INSTITUTIONAL FRAMEWORK, THE STYLIZED BALANCE SHEET AND THE EVOLUTION OF CRYPTOASSET USE

We consider a model economy populated by five distinct entities: central banks; commercial banks; NBFIs; households; and stablecoin issuers. In this section we detail the six-stage stylized process used to construct our multi-entity balance sheet framework incorporating all four asset classes, describing the balance sheet implications of the issuance of stablecoins and CBDCs, their availability and their use by market participants.

3.1 Stage 1: baseline without digital assets

We begin by assuming that in stage 1 there are no cryptocurrencies, stablecoins or CBDCs. Table 1 describes this baseline scenario.

3.2 Stage 2: cryptocurrency adoption as medium of exchange and bank disintermediation

As detailed above, the financial system in stage 1 comprised only the central bank, commercial banks, NBFIs and households. In the second stage the introduction of cryptocurrencies created a new channel for potential bank disintermediation, as illustrated in Table 2. With the publication of the foundational paper by Nakamoto (2008), Bitcoin emerged as a potential medium of exchange within the economy. At this

TABLE 1 Baseline balance sheet configuration prior to digital asset adoption (stage 1).

(a) Central banks	
Assets	Liabilities
Treasury securities	Reserves Fiat money
(b) Commercial banks	
Assets	Liabilities
Cash Reserves Asset securities Treasury securities Loans	Bank deposits
(c) Households	
Assets	Liabilities
Cash Bank deposits Treasury securities MMFs	Loans
(d) NBFIs	
Assets	Liabilities
Treasury securities Commercial papers Wholesale bank deposits Repos Asset securities Foreign currencies	Mutual fund shares

time, cryptocurrencies were primarily held by households, who began substituting their bank deposits with cryptocurrency. This shift occurred as Bitcoin introduced a new form of money, distinct from traditional currencies.²

² On May 22, 2010, the first Bitcoin transaction occurred, when developer Laszlo Hanyecz used 10 000 bitcoins (BTC), which was worth US\$41 at that time, to purchase two pizzas (Montgomery 2025).

TABLE 2 Balance sheet impact of cryptocurrency adoption as a medium of exchange (stage 2).

(a) Central banks	
Assets	Liabilities
Treasury securities	Reserves Fiat money
(b) Commercial banks	
Assets	Liabilities
Cash Reserves ↓ Treasury securities Asset securities Loans	Bank deposits ↓
(c) Households	
Assets	Liabilities
Cash Bank deposits ↓ Cryptocurrency ↑ Treasury securities MMFs	Loans

When households purchase cryptocurrencies through centralized exchanges such as Binance and Coinbase, they typically fund these transactions with their commercial bank deposits. This action triggers a direct outflow from the banking system, causing a contraction in the commercial bank’s balance sheet. Specifically, the bank experiences a simultaneous decrease in its deposit liabilities and, critically, its reserve assets. This erosion of its core funding base tightens the bank’s liquidity. For the household, however, this is simply a portfolio reallocation; the overall size of their balance sheet remains unchanged as one asset (deposits) is substituted for another (cryptocurrencies). For the NBFI, the balance sheet remains unchanged.

3.3 Stage 3: cryptocurrency as a speculative asset

In stage 3, households begin to incorporate Bitcoin into their investment portfolios, not as a substitute for cash but as a substitute for other traditional financial assets.

TABLE 3 Balance sheet shifts when cryptocurrency transitions to a speculative asset (stage 3).

(a) Households	
Assets	Liabilities
Cash ↓ Bank deposits ↓ Cryptocurrency ↑ Treasury securities ↓ MMFs ↓	Loans
(b) Commercial banks	
Assets	Liabilities
Cash ↓ Reserves ↓ Treasury securities Asset securities Loans	Bank deposits ↓
(c) NBFIs	
Assets	Liabilities
Treasury securities ↓ Commercial papers Wholesale bank deposits ↓ Repos Asset securities Foreign currencies	Mutual fund shares ↓

This reflects the evolving narrative surrounding cryptocurrency. Due to its high price volatility (Wu *et al* 2025), its function as a stable medium of exchange diminishes, and it becomes widely viewed as a speculative financial asset.

As shown in Table 3, a common reallocation strategy involves households not only reducing their bank deposits but also selling safer assets, such as Treasury securities and MMFs, to fund their crypto purchases in pursuit of higher returns. This reallocation not only drains retail deposits from commercial banks but also draws liquidity away from NBFIs. It represents a significant shift in risk appetite.³

³ This shift has been facilitated first by technology improvements that have provided easy access to financial markets and crypto exchanges, and second by fractional trading, which has allowed households to trade fractions of regular stock shares.

TABLE 4 Balance sheet composition of three stablecoin issuer types by reserve structure (stage 4).

(a) Fiat-backed stablecoin issuers	
Assets	Liabilities
Bank deposits Short-term Treasuries, repos MMFs	Fiat-backed stablecoin
(b) Crypto-backed stablecoin issuers	
Assets	Liabilities
Fiat-backed stablecoins Cryptocurrencies (BTC, ETH, etc) Tokenized Treasuries	Crypto-backed stablecoin
(c) Algorithmic stablecoin issuers	
Assets	Liabilities
Governance token	Algorithmic stablecoin

More importantly, this portfolio reallocation continues to exert pressure on the commercial banking system. To manage these deposit outflows and maintain central bank reserve requirements, banks initially draw down their cash and excess reserves. However, if the withdrawal pressure persists, commercial banks are forced to liquidate Treasury holdings and other asset securities. This forced deleveraging acts as a critical transmission channel, potentially creating systemic liquidity issues for broader financial and real markets. Further, although the transaction takes place between two financial assets, the settlement process (selling Treasuries and subsequently buying crypto) flows through the banking system, causing interbank transfers of reserves. Therefore, similarly to stage 2, these activities can create liquidity fluctuations and tighten funding conditions for individual banks, challenging the notion that bank balance sheets remain unaffected.

3.4 Stage 4: the introduction of stablecoins and banking interlinkages

In stage 4, a new entity is introduced into the economy: the stablecoin issuer. These issuers create digital tokens pegged to traditional fiat currencies or other financial

instruments to provide stability in the cryptocurrency ecosystem, adding a new layer of complexity to the financial landscape.

Stablecoins were introduced with the issuance of the Tether stablecoin (USDT) on October 6, 2014, with the aim of stabilizing the value of digital tokens. Stablecoins can be broadly categorized by their reserve structure, which dictates their balance sheet composition and their impact on the traditional financial system (see Table 4).

Fiat-backed stablecoins. These tokens, such as USDT or USDC, are backed by reserves of cash and cash equivalents.⁴ More importantly, rather than holding only bank deposits, major issuers allocate substantial portions of their fiat backing into government MMFs. This structural choice further shifts liquidity away from traditional commercial banks and toward NBFIs.

Crypto-backed stablecoins. A distinct class of stablecoins, exemplified by USDS (the stablecoin of the Sky Protocol ecosystem), employs an overcollateralized lending mechanism rather than direct fiat backing. Users deposit cryptoassets such as Ethereum (ETH), BTC or even fiat-backed stablecoins such as USDC into smart contract vaults, and they mint stablecoins against this collateral at a collateralization ratio typically exceeding 150%. This structural difference has several implications. First, for the user, minting crypto-backed stablecoins is economically equivalent to a collateralized loan: the user retains price exposure to the underlying cryptoasset while extracting US-dollar-denominated liquidity. This leverage function has no counterpart in fiat-backed stablecoins, where the user surrenders the dollar entirely. Second, because the protocol is governed by a decentralized autonomous organization (DAO) rather than a corporate issuer, no single entity can freeze or blacklist token holdings, particularly after Circle froze USDC addresses associated with sanctioned entities in 2022. Third, crypto-backed stablecoins generate endogenous yield through stability fees charged to borrowers, which can be distributed to token holders, whereas fiat-backed stablecoin yields accrue to the issuer. However, this architecture introduces a critical vulnerability: procyclical liquidation risk. When collateral values decline sharply, the protocol automatically liquidates undercollateralized positions, forcing sales of the underlying cryptoassets. These forced sales can depress prices further, triggering additional liquidations, a feedback loop demonstrated during the

⁴ These broadly include highly liquid, short-term investments such as US Treasury bills, MMFs, reverse repurchase agreements, highly rated commercial paper and bank deposits. These assets provide yield while maintaining the immediate liquidity required to facilitate 1:1 token redemption (when an issuer receives US\$1, it mints one token).

March 2020 “Black Thursday” event, when ETH prices fell 43% in 24 hours and MakerDAO suffered US\$8.32 million in undercollateralized liquidations (Coutts 2020).

The collateral composition of major crypto-backed stablecoins has evolved significantly. MakerDAO’s reserves now include substantial allocations to tokenized US Treasuries and other RWA alongside traditional crypto collateral. This evolution creates a direct transmission channel between the RWA market and the stablecoin ecosystem: a default or devaluation in tokenized RWA collateral could impair the solvency of the stablecoin protocol, a risk we analyze further in our discussion of DeFi composability (see stage 5.1).

Algorithmic stablecoins. A third class of stablecoins, exemplified by the now-collapsed Terra USD (UST), relies on algorithmic protocols rather than reserves. In a two-coin system, a governance token (eg, Luna) is minted or burned to absorb volatility and maintain the stablecoin’s peg. The issuer’s balance sheet consists only of these two interconnected tokens. The minting of algorithmic stablecoins has no direct impact on the banking system’s balance sheets, as their mechanism is self-contained and does not rely on external reserves held in banks. This distinction is critical for understanding the differential risk profiles of various stablecoin models.

In what follows we focus on fiat- and crypto-backed stablecoins (Table 5). The interconnectedness between these two types of stablecoins creates a powerful transmission channel back to the traditional banking system. When a crypto-backed stablecoin issuer mints new tokens, it must acquire more collateral, often in the form of fiat-backed stablecoins. This forces the fiat-backed issuer to mint more tokens, for which it must acquire more US-dollar-denominated reserves. A significant portion of these reserves is held as large corporate deposits within commercial banks (part (a)).⁵ This directly expands the commercial bank’s balance sheet by increasing both its deposit liabilities and, critically, its reserve assets.

The structural reliance of fiat-backed stablecoins on MMFs creates a significant spillover effect on the nonbanking sector. As shown in the NBF balance sheet (part (b)), NBFs experience passive balance sheet expansion. Stablecoin issuers effectively act as aggregators: they sweep retail liquidity out of the traditional

⁵ Fiat-backed stablecoin issuers now place most of their bank deposits with systemically important banks in order to mitigate counterparty risk, a key lesson learned from the 2023 Silicon Valley Bank bankruptcy.

TABLE 5 Commercial bank and NBFIs balance sheet expansion driven by fiat-backed stablecoin reserve inflows (stage 4).

(a) Commercial banks	
Assets	Liabilities
Cash ↑	Bank deposits ↑
Reserves ↑	
Treasury securities	
Asset securities	
Loans	
(b) NBFIs	
Assets	Liabilities
Treasury securities ↑	MMFs (held by stablecoin issuers) ↑
Commercial papers	MMFs (held by households)
Wholesale bank deposits	
Repos ↑	
Asset securities	
Foreign currencies	

commercial banking system and channel it into NBFIs.⁶ Consequently, on the liability side of the NBFIs balance sheet, there is a substantial increase in MMF shares held by these institutional stablecoin issuers. To absorb this inflow, NBFIs proportionally expand their asset holdings in short-term government securities and repurchase (repo) markets. These dynamics underscore a critical structural shift: the proliferation of stablecoins has accelerated the migration of credit intermediation from traditional commercial banks to the nonbank financial sector.

⁶ As of March 2026, most of Circle’s USDC reserves (US\$75.9 billion in total) were held through the Circle Reserve Fund (USDXX), an SEC-registered Rule 2a-7 government MMF managed by BlackRock – the fund’s portfolio consisted of overnight US Treasury repurchase agreements (58.5% of total reserves) and short-dated US Treasuries (25.6%), with the remainder held as cash deposits at systemically important financial institutions (Circle Internet Group, Inc. 2026) – while Tether reported direct and indirect exposure to US Treasury bills of approximately US\$141 billion (the indirect portion held through money market funds and repurchase agreements) out of total assets of US\$191.8 billion, with the remainder allocated to physical gold, Bitcoin, secured loans and other investments (BDO Advisory Services 2026).

3.5 Stage 5: commercial bank balance sheet with active participation

In stage 5, the financial landscape undergoes a structural transformation as traditional intermediaries shift from passive infrastructure providers to active participants in the digital asset ecosystem. This development signals a shift in the financial landscape, where commercial banks begin to engage in two distinct novel activities: acting as stablecoin issuers and holding cryptoassets as principal investments (Heeb and Baer 2025).

When a bank issues its own fiat-backed stablecoin, it primarily triggers a liability-side transformation. A customer typically purchases the bank-issued stablecoin using their existing deposits at that same institution. This results in a direct swap on the bank's balance sheet: deposit liabilities decrease while stablecoin liabilities increase. The bank's total funding base may initially remain stable, but its composition and risk profile change significantly.

These dual roles fundamentally recompose the bank's balance sheet. To fund its new digital asset holdings and stablecoin issuance, the bank must reallocate resources from its existing asset portfolio. Three scenarios illustrate the range of possible adjustments, each with distinct implications for the bank's risk profile and its role in the broader economy.

Scenario A: lending contraction. The bank reduces its loan book to free up capital for crypto investments. This is the most damaging scenario from a macroeconomic perspective: every dollar diverted from lending to crypto represents a direct reduction in credit available to households and firms. The bank's core intermediation function is diminished.

Scenario B: securities liquidation. The bank sells Treasury securities and/or asset-backed securities from its investment portfolio. While this preserves lending capacity, it reduces the bank's liquidity buffer and its holdings of high-quality liquid assets, potentially impairing its ability to meet regulatory liquidity requirements such as the liquidity coverage ratio and net stable funding ratio (NSFR) and weakening its resilience to future stress events.

Scenario C: mixed reallocation. In practice, banks are likely to pursue a combination of both strategies, simultaneously reducing loans, Treasuries, asset securities and even cash holdings (Table 6). This blended approach distributes the adjustment across the entire asset side, but the cumulative effect is a fundamental shift in the bank's asset composition: from safe, liquid and regulatorily favored assets toward volatile, illiquid and regulatorily uncertain digital assets.

TABLE 6 Commercial bank balance sheet recomposition under active digital asset participation (stage 5).

Assets	Liabilities
Cash ↓	Bank deposits ↓
Reserves	Stablecoins issued ↑
Treasury securities ↓	
Asset securities ↓	
Loans ↓	
Cryptocurrencies (as investment) ↑	
Stablecoins (held as asset) ↑	

TABLE 7 Commercial bank balance sheet as custodian for stablecoin reserve assets (stage 5).

Assets	Liabilities
Cash	Bank deposits
Reserves ↑	Stablecoins issued
Treasury securities	Custody deposits (fiat reserves) ↑
Asset securities	
Loans	
Cryptocurrencies (as investment)	
Stablecoins (held as asset)	

| **Memorandum items (off-balance sheet)** | |
| Custodial digital assets and securities | |

Further, commercial banks could increasingly assume the role of custodians for stablecoin issuers. In this capacity, banks safeguard the reserve assets that back fiat-backed stablecoins.

In this case, the accounting treatment differs based on the asset type. When a stablecoin issuer deposits cash reserves, it acts as a massive inflow of corporate deposits. As shown in Table 7, this expands both sides of the bank’s balance sheet: the bank gains reserves and incurs a liability in the form of custody deposits. Conversely, if the bank holds securities or digital assets in segregated accounts on behalf of the issuer, these are typically recorded as off-balance-sheet items, generating fee income rather than altering the balance sheet structure.

3.6 Stage 5.1: the tokenization of RWA

The balance sheet implications of RWA tokenization differ substantially depending on the nature of the underlying asset. We distinguish two cases: financial RWA (such as tokenized Treasuries, corporate bonds and private credit) and nonfinancial RWA (such as tokenized real estate and commodities). The distinction matters because the two cases produce fundamentally different effects on the banking system's funding base and credit intermediation capacity.

While the assets introduced in stages 2 and 3 are “digitally native” (Bitcoin and Ether exist only as ledger entries; the token itself constitutes the asset and no off-chain counterpart exists), RWA tokens are “digital twins”: on-chain representations of assets (Treasury bills, private loans, gold bars, buildings) that exist, and remain, off-chain (Organisation for Economic Co-operation and Development 2020). A twin has no independent value: the holder's rights derive from an off-chain legal claim that must be created, custodied and enforced through traditional institutions, so every twin requires a bridge between the ledger and the conventional legal system (Aldasoro *et al* 2023). Fiat-backed stablecoins occupy an intermediate position (newly created on-chain liabilities backed by a reserve pool, rather than one-to-one representations of preexisting instruments), which is why their balance sheet mechanics (stage 4) differ from those analyzed here. The twin structure has two consequences.

- (i) Some entity must hold the underlying asset and anchor the legal claim; this is the role of the special purpose vehicle (SPV), and it is why the analysis below must span four balance sheets: the investor, the SPV, the investor's bank and the SPV's custodian bank.
- (ii) The token transfers at blockchain speed, around the clock, while the underlying claim can be enforced only at the speed of custodians, registrars and courts. This asymmetry between on-chain transferability and off-chain enforceability underlies the recurring redemption and enforcement risks, which we illustrate in two cases below.

Because the twin's legal claim must be anchored off-chain, the issuance of RWA tokens typically relies on an SPV structure. The SPV acquires underlying assets, issues digital tokens on the blockchain representing fractionalized claims on its asset pool and acts as a pass-through entity: on the asset side it holds the collateral (often in custody at regulated financial institutions), while on the liability side it issues tokens that circulate on-chain, representing investors' pro rata rights to the underlying cashflows or liquidation value.

3.6.1 Case 1: tokenized financial assets

The largest segment of the RWA market consists of tokenized financial instruments. As of February 2026, tokenized US Treasuries (eg, BlackRock's BUIDL, Circle's USYC, Ondo's USDY), tokenized corporate bonds (eg, Blockstream's BMN2) and tokenized private credit (eg, Syrup USDC) collectively accounted for most nonstablecoin RWA market capitalization.⁷

3.6.1.1 Funding flow and balance sheet mechanics. The complete transaction chain begins with the household (or institutional investor). Consider a household that uses bank deposits to purchase RWA tokens. The funds flow through the following sequence.⁸

- (1) The household withdraws deposits from its commercial bank to purchase RWA tokens from the SPV.
- (2) The SPV receives the cash and deploys it into the target financial assets: short-term Treasuries, money market instruments or private credit facilities.
- (3) A portion of the SPV's cash may remain as operational reserves at a custodian bank; the remainder exits the banking system entirely as it flows into capital markets.

Table 8 traces the balance sheet impact across all four affected entities. Several observations can be made from this table. First, for the household, this is a straightforward asset substitution: bank deposits are exchanged for RWA tokens, leaving total wealth unchanged. Second, the net effect on the aggregate banking system depends on how much of the SPV's incoming cash is retained as bank deposits versus the amount deployed into nonbank financial markets. If the household's bank and the SPV's custodian bank are different institutions, the transaction creates a redistribution of reserves and deposits across banks: one bank contracts while another expands. In aggregate, however, the portion of funds that the SPV channels into Treasuries or private credit markets represents a net leakage from the banking system's deposit base, structurally analogous to the liquidity migration observed in stage 4 when stablecoin issuers allocate reserves to MMFs.

⁷ URL: <https://app.rwa.xyz>.

⁸ We focus on primary-market subscriptions, whether executed directly with the SPV or via a distribution platform; secondary-market transfers between investors merely redistribute deposits across banks and generate no new SPV inflows.

TABLE 8 Multi-entity balance sheet impact of tokenized financial asset issuance through an SPV (stage 5.1, case 1).

(a) Household	
Assets	Liabilities
Cash	Loans
Bank deposits ↓	
Cryptocurrency	
Treasury securities	
MMFs	
RWA tokens ↑	
(b) SPV: tokenized Treasuries/private credit	
Assets	Liabilities
Short-term Treasuries	RWA tokens (issued)
Private credit/loans	
Cash reserves (at custodian)	
(c) Commercial bank (household's bank)	
Assets	Liabilities
Cash	Bank deposits ↓ (household withdrawal)
Reserves ↓	
Treasury securities	
Asset securities	
Loans	
(d) Custodian bank (SPV's bank)	
Assets	Liabilities
Reserves ↑	Bank deposits ↑ (SPV cash reserves)

3.6.1.2 Shadow credit intermediation. The case of tokenized private credit deserves particular attention. When an SPV uses investor funds to originate loans (whether for small and medium-sized enterprise lending, trade finance or real estate debt), it is performing the core function of a bank (credit intermediation) without being subject to banking regulation. The SPV operates outside the perimeter of reserve requirements, capital adequacy ratios and deposit insurance. This creates a “shadow

lending channel”: credit is created in the economy, but it does not appear on any regulated bank’s balance sheet. The systemic risk implications are significant: a wave of defaults in tokenized private credit pools would not trigger the regulatory safety nets, such as Federal Deposit Insurance Corporation (FDIC) deposit insurance, that protect traditional bank lending.

3.6.2 Case 2: tokenized nonfinancial assets

In the second case, the SPV tokenizes physical or tangible assets such as gold (Tether Gold, Paxos Gold), energy (JWMH) or real estate (Reental). The balance sheet mechanics differ from case 1 in a critical respect: the underlying assets are “illiquid and nonfinancial”, and the custodian bank’s role is limited to safekeeping rather than intermediation. Case 2 is the digital twin in its purest form: the underlying asset cannot itself move on-chain, so the token’s value rests entirely on the integrity of the custody arrangement.

As shown in Table 9, the custodian bank’s balance sheet is largely unaffected. The physical assets are held in segregated custody accounts and recorded as off-balance-sheet memorandum items, generating fee income for the bank without altering its leverage or liquidity ratios. The primary balance sheet impact falls on the household and the household’s originating bank. However, since the SPV’s cash holdings are typically small (the bulk of the raised capital is used to acquire the physical asset), the net deposit leakage from the banking system is less severe than in case 1.

3.6.3 DeFi composability and cross-asset transmission

Regardless of the underlying asset type, once RWA tokens are issued on a permissionless blockchain, they become subject to the DeFi composability dynamics discussed in stage 5.2, case B. The most prominent example is MakerDAO (now Sky), which accepts tokenized Treasuries and private credit as collateral to mint the DAI stablecoin. This creates a cross-asset transmission chain with significant systemic implications.

- (1) The household purchases tokenized Treasury tokens from an SPV (stage 5.1, case 1).
- (2) The household deposits those RWA tokens into a DeFi lending protocol as collateral.⁹

⁹ The legal characterization of step (2) matters more than its mechanics. Singh (2011) distinguishes a pledge (under which the collateral taker acquires no right of reuse unless one is expressly granted in writing) from a title transfer (under which ownership passes and rights of reuse are inherent in the arrangement), reserving “rehypothecation” for onward pledging by the taker as security for its own obligations and using “reuse” for any dealing consistent with ownership. Posting an RWA

TABLE 9 Balance sheet impact of tokenized nonfinancial assets with off-balance-sheet custody (stage 5.1, case 2).

(a) Households	
Assets	Liabilities
Cash	Loans
Bank deposits ↓	
Cryptocurrency	
Treasury securities	
MMFs	
RWA tokens ↑	
(b) SPV: tokenized real estate/commodities	
Assets	Liabilities
Commodities	RWA tokens (issued)
Cash (operational)	
(c) Custodian bank	
Assets	Liabilities
Minimal on-balance-sheet impact	Minimal on-balance-sheet impact
Memorandum items (off-balance sheet)	
Custodial real estate deeds/commodity certificates	

- (3) The protocol mints stablecoins against the RWA collateral.
- (4) The newly minted stablecoins circulate in the DeFi ecosystem, effectively creating new purchasing power backed by the original Treasury holdings.

token to a lending protocol is a title transfer in all but name: the token moves to a contract address, the protocol acquires unconditional control, and no agreement confers or limits what may be done with it. Reuse rights arise by construction rather than by negotiation, which is why the layering described below requires no counterparty consent at any step. The characterization also governs default. A title transfer entitles the provider to equivalent collateral rather than the original asset, and in practice remedies against an insolvent taker are extremely limited: a smart contract has neither legal personality nor an insolvency estate, so the provider’s remedy is not merely limited but undefined.

This chain generates a “DeFi credit multiplier”: a single dollar of bank deposits, once it has been converted into an RWA token and reused through DeFi, can support multiple layers of on-chain credit, none of which appear on any bank’s balance sheet. Table 10 illustrates the balance sheet of a DeFi participant who rehypothecates RWA tokens.

A stress event at any point in the chain (eg, a depegging of the stablecoin, a smart contract exploit or a sudden drop in the value of the underlying RWA) could trigger cascading liquidations that propagate back to the traditional financial system through redemption pressures on the SPV and, ultimately, on the custodian bank’s reserves. Figure 2 traces the chain from the original bank deposit through to the replighted stablecoins, together with the stress path that runs back along it.

Lee (2026) formalizes the internal fragility of such cross-asset transmission chains: in an economy of vertically layered protocols, upper layers broaden user access but bootstrap contingent liquidity from the layers beneath them; the base protocol responds by rationing liquid reserves; and composability generates a waterfall of externalities that, under severe conditions, reduces their total utilization. Lee’s analysis is complementary to ours: it asks what composability does to liquidity provision within the protocol economy, taking the tokenized instruments as given, while we seek to discover where the exposures originate and where they land when the chain unwinds. The layering in Figure 2 begins with a deposit on a regulated bank’s balance sheet and ends, when stress runs in reverse, at the custodian bank’s reserve position; neither endpoint is visible in a model confined to the protocol economy. Lee (2026) supplies the reason the upper layers are structurally illiquid, while the accounting here identifies the regulated institutions that absorb that illiquidity.

This cross-asset transmission, from bank deposits to RWA tokens to DeFi-minted stablecoins, is where tokenization amplifies systemic leverage beyond the reach of prudential regulation, and it is the point at which the on-chain and regulated balance sheets are combined.

3.7 Stage 5.2: bank-issued digital assets

This section discusses the commercial bank’s transition into a primary issuer of digital liabilities: the most significant evolution after custody and intermediation. This transition took two distinct forms: the immediate adoption of tokenized deposits such as JPM Coin; and the prospective issuance of digital securities (eg, digital bonds). Critically, the balance sheet implications of these instruments depend not only on their legal classification but also on the infrastructure of the underlying blockchain on which they are issued. We therefore analyze two cases: issuance on a permissioned blockchain versus issuance on a permissionless blockchain.

FIGURE 2 The DeFi credit multiplier: US\$1 of bank deposits supporting successive layers of on-chain credit.

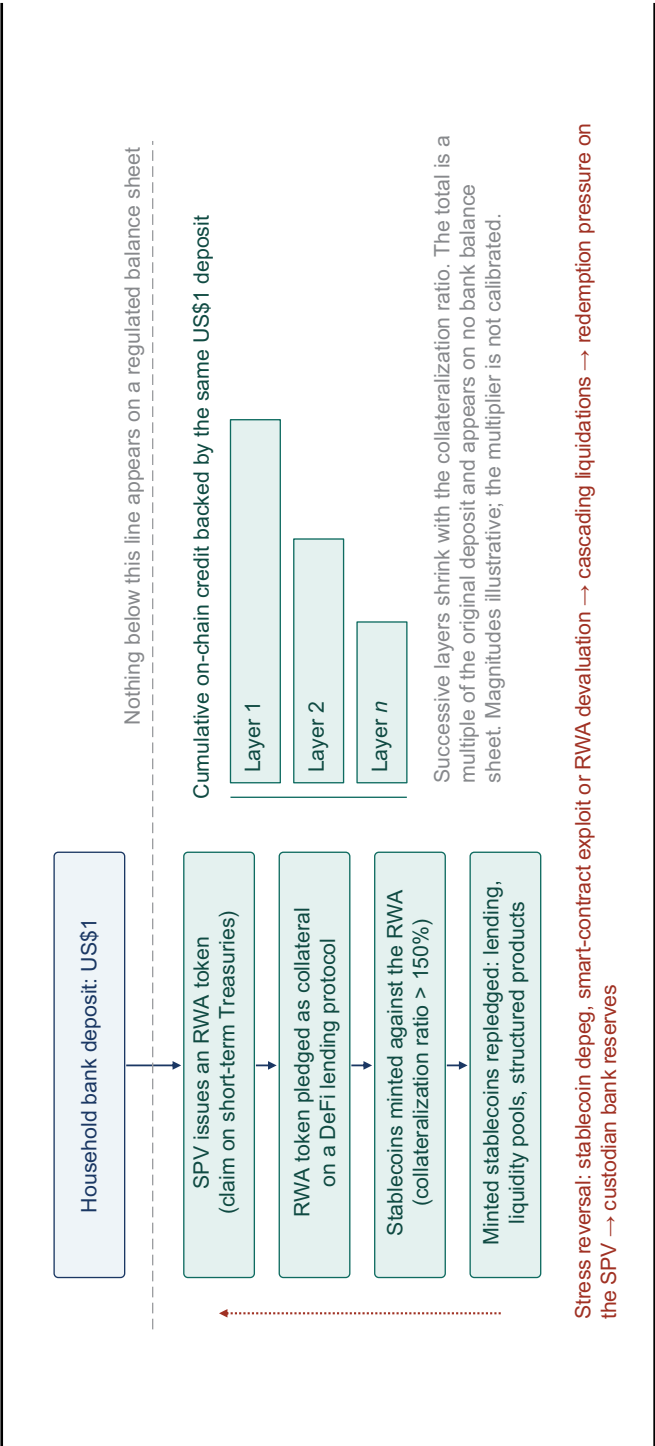


TABLE 10 Balance sheet of a DeFi participant (household/fund) rehypothecating RWA tokens.

Assets	Liabilities
RWA tokens (as collateral)	DeFi borrowings (stablecoins) ↑
Stablecoins (borrowed) ↑	
Other DeFi positions	

TABLE 11 Liability substitution and balance sheet expansion from bank-issued digital assets on a permissioned blockchain (stage 5.2, case A).

(a) Tokenized deposits: permissioned (liability substitution)	
Assets	Liabilities
Reserves	Legacy deposits ↓
Loans	Tokenized deposits ↑
Securities	(Deposit insurance applies)

(b) Digital bonds: permissioned (balance sheet expansion)	
Assets	Liabilities
Reserves ↑ (proceeds)	Bank deposits
Loans	Digital bonds ↑

3.7.1 Case A: issuance on a permissioned blockchain

In the permissioned model, exemplified by JPM Coin, the bank deploys tokens on a private, access-controlled distributed ledger. Only counterparties that have undergone know-your-customer/anti-money-laundering verification and have been explicitly whitelisted by the bank can hold, transfer or redeem the tokens. The bank retains full administrative authority over the network: it can freeze accounts, impose transfer limits, enforce circuit breakers and unilaterally revoke access.

Under this architecture, the issuance of tokenized deposits is fundamentally a liability-side reclassification. When a client converts traditional funds into tokenized deposits, the bank simply relabels the liability. As shown in Table 11, the total size of the balance sheet remains unchanged; the composition of liabilities shifts from “legacy deposits” to “tokenized deposits”, but both remain deposit liabilities under the same regulatory regime. Deposit insurance (eg, FDIC deposit insurance in the United States) continues to apply, reserve requirements are unaffected and

the bank's liquidity risk profile is essentially preserved. The token cannot leave the permissioned environment and therefore cannot be reused or repurposed in external markets.

Similarly, the issuance of digital bonds on a permissioned ledger follows the mechanics of traditional bond issuance. The bank receives reserves on the asset side and records a new long-term debt obligation on the liability side. The primary benefit is operational (faster settlement, programmable coupon payments) rather than structural.

This regulatory continuity is now codified in the Basel Framework. Under the Basel Committee's cryptoasset standard (SCO60),¹⁰ in force since January 1, 2026, tokenized versions of traditional assets that satisfy the classification conditions (including legal enforceability of the holder's claim, settlement finality, and the requirement that all entities executing transfers, settlement and redemptions be regulated and supervised) are assigned to Group 1a and receive capital requirements based on the risk weights of their conventional counterparts (Basel Committee on Banking Supervision 2022, 2024; Ye 2025).¹¹

The key feature of the permissioned model is that the bank's digital liabilities remain trapped within a walled garden. The token is merely a more efficient representation of an existing claim, and the bank retains the same degree of control over its funding base as in the traditional system. In this sense, permissioned tokenization reinforces, rather than disrupts, the existing banking architecture.

A qualification is necessary: regulatory continuity is not the same as visibility. Singh (2026) observes that tokenized repo on bank-operated platforms typically opens and closes within the same day, so the end-of-day snapshot on which balance sheet reporting rests records a zero position and the intraday reuse appears neither on the balance sheet nor in its footnotes. The walled garden that preserves the prudential treatment of the token also encloses the record of how often that token moved. The permissioned model therefore reinforces the existing architecture along its legal and capital dimensions while eroding the one input on which supervision of collateral reuse depends: a measurable record that the reuse occurred.

¹⁰ The standard (Chapter SCO60 of the consolidated Basel Framework), effective from January 1, 2026, was finalized in December 2022 and amended in July 2024. National implementation varies: the European Union has partially transposed the standard through CRR3, while US implementation remains pending.

¹¹ Permissioned issuance is, in effect, engineered to satisfy these conditions: the issuing bank controls validator access, guarantees settlement finality and remains the regulated entity responsible for redemption. For a counterparty bank holding the tokenized deposit or digital bond, the capital charge therefore mirrors that of the equivalent traditional claim, subject at most to an infrastructure risk add-on that national authorities may activate at their discretion.

3.7.2 Case B: issuance on a permissionless blockchain

The balance sheet dynamics change fundamentally when a bank issues tokens on a permissionless blockchain such as Ethereum. In this environment, tokens are bearer instruments: once issued, they circulate freely among any wallet address without the bank's authorization or oversight. The bank cannot freeze, restrict or recall tokens that have been transferred to third parties. This architectural difference has three critical implications for the bank's balance sheet and the broader financial system, which we discuss below.

3.7.2.1 Liability reclassification and regulatory ambiguity. Unlike tokenized deposits on a permissioned ledger, tokens issued on a public blockchain may no longer qualify as "deposits" under existing regulatory frameworks. Once a token can be held by anonymous, unverified parties and traded on secondary markets, including decentralized exchanges (DEXs), it acquires the characteristics of e-money or even a security, depending on its jurisdictional classification. This reclassification has direct consequences on both sides of the balance sheet. On the liability side, the instrument may no longer be covered by deposit insurance and may not count toward the bank's stable funding in NSFR calculations. On the asset side (for any regulated bank holding the token), the consequences are starker. Under SCO60, the Basel Committee's current position is that the risks arising from permissionless blockchains cannot be sufficiently mitigated to permit Group 1 classification; tokens issued on such chains therefore fall into Group 2, where unhedged exposures carry a 1250% risk weight and a bank's aggregate Group 2 holdings are expected to remain below 1% of Tier 1 capital, with a hard limit of 2% (Basel Committee on Banking Supervision 2022, 2024).¹² The practical effect is not to eliminate the exposure but to displace it: holdings that are prohibitively expensive for regulated banks migrate to unregulated intermediaries and DeFi protocols: precisely the environment analyzed in Section 3.7.2.3. As shown in Table 12, we denote this liability "digital token liabilities" to distinguish it from regulated deposits.

3.7.2.2 Amplified run risk. On a permissioned ledger, the bank can impose redemption gates and transfer delays, providing a buffer against rapid outflows. On a permissionless chain, no such safeguard exists. Token holders can instantaneously swap bank-issued tokens for USDC, ETH or any other asset on a DEX, bypassing the bank entirely. This transforms the speed of potential bank runs from days to seconds. The commercial bank faces a structurally higher liquidity risk: it must maintain sufficient

¹² In November 2025 the Basel Committee launched a targeted review of the standard, including the treatment of permissionless blockchains, with an update expected in the course of 2026.

TABLE 12 Balance sheet and off-balance-sheet exposures from bank-issued tokens on a permissionless blockchain (stage 5.2, case B).

(a) Commercial bank: permissionless issuance	
Assets	Liabilities
Cash	Legacy deposits ↓
Reserves	Digital token liabilities ↑
Treasury securities	(Deposit insurance may not apply)
Asset securities	
Loans	
Memorandum items (off-balance sheet)	
DeFi reuse of bank-issued tokens	
Liquidity pool exposures through DEXs	
Smart contract counterparty risk	
(b) Household/DeFi participant	
Assets	Liabilities
Bank-issued tokens	DeFi borrowings ↑
DeFi lending positions ↑	
LP tokens ↑	

reserves to meet redemptions that can occur at blockchain speed (around 12 seconds per block on Ethereum L1), without the ability to invoke traditional emergency mechanisms such as suspension of convertibility.

3.7.2.3 Off-balance-sheet leverage via DeFi composability. Perhaps the most profound distinction from issuance on a permissioned blockchain is the composability of tokens. Once bank-issued tokens enter the DeFi ecosystem, they can be

- deposited as collateral on lending protocols (eg, Aave, Compound) to borrow other assets;
- supplied as liquidity to automated market makers (eg, Uniswap, Curve); and
- wrapped, stacked or bundled into structured products by third-party smart contracts.

Each of these activities effectively creates a shadow credit multiplier: the bank’s original liability is reused multiple times across DeFi protocols, generating new layers

of credit that exist entirely outside the banking system's balance sheet. This is analogous to the pre-2008 shadow banking system, where mortgage-backed securities were repeatedly pledged and repledged through repo chains, except that the process is now automated, permissionless and operates 24/7 without any central clearing counterparty. This comparison understates the difference in one respect: repledging in the United States is capped by Regulation T and SEC Rule 15c3-3, limits stringent enough that leverage-seeking strategies have migrated offshore (Singh 2011).¹³

From the bank's perspective, these activities are invisible: they do not appear on its balance sheet, yet they create systemic exposures that are ultimately linked back to the bank's creditworthiness. We capture this in Table 12(a) as an off-balance-sheet memorandum item.

The household/DeFi participant T-account in Table 12(b) illustrates the downstream effects. A household that receives bank-issued tokens on a public chain can immediately deploy them as collateral to borrow additional assets, creating leveraged positions that amplify both individual and systemic risk. The "LP tokens" represent claims on liquidity pool positions in DEXs, which themselves carry impermanent loss and smart contract risk.¹⁴ None of these derivative exposures are visible on the originating bank's balance sheet, yet a depegging event or smart contract failure could trigger a cascade of liquidations that ultimately result in mass redemptions of the bank's tokens.

3.7.3 *Issuance on a permissioned versus a permissionless blockchain*

Table 13 synthesizes the structural differences between the two blockchain architectures from the perspective of the issuing bank. This distinction carries a fundamental insight: the choice of blockchain infrastructure determines whether bank-issued digital assets reinforce the existing financial architecture or catalyze a new generation of shadow banking. On a permissioned chain, tokenization is an efficiency gain that preserves the bank's role as a controlled intermediary. On a permissionless chain, the same instrument becomes a vector for regulatory arbitrage and for leverage accumulation outside the prudential perimeter, structurally analogous to the off-balance-sheet conduits and structured investment vehicles that amplified the 2007–9 global financial crisis.

¹³ In August 2026, the SEC proposed Regulation Crypto Assets, introducing a tailored securities offering framework for certain cryptoasset investment contracts, including registration exemptions and a conditional safe harbor from classification as a security. The proposal is also relevant to Regulation T, as its clarification of when cryptoassets cease to constitute securities may determine whether broker-dealer credit involving such assets is subject to Regulation T's margin requirements (US Securities and Exchange Commission 2026b).

¹⁴ Impermanent loss refers to the unrealized loss that a liquidity provider incurs when the relative price of the two assets in a pool diverges from the price at the time of deposit.

TABLE 13 Structural comparison of permissioned versus permissionless blockchain issuance.

Dimension	Permissioned	Permissionless
Liability classification	Deposit (regulated)	Digital token (regulatory gray area)
Deposit insurance	Applies	May not apply
Prudential treatment	Group 1a eligible; capital mirrors traditional assets	Group 2; up to 1250% risk weight, 1%–2% Tier 1 exposure cap
Bank control	Full (freeze, restrict, revoke)	None post-issuance
Run velocity	Gated (bank-controlled)	Instantaneous (DEX-enabled)
Secondary market	Closed (whitelisted only)	Open (any wallet)
Collateral reuse	Pledge-like: no reuse without express agreement	Title-transfer-like: reuse inherent, no agreement needed
Observability of reuse	Intraday activity nets to zero in end-of-day reporting	Every transfer is a public ledger
Off-balance-sheet exposure	None	Shadow credit multiplier via DeFi
Net balance sheet effect	Neutral (liability substitution)	Neutral on-balance-sheet, but systemic leverage ↑ off-balance-sheet
Analogy	Internal ledger upgrade	New shadow banking channel

The analogy breaks in one place, and the break is instructive. The conduits of 2007 were invisible to everyone outside the sponsoring bank, whereas a permissionless chain records every transfer it processes. Set against the intraday opacity of the bank-operated platforms, this yields an inversion worth stating plainly: the venue that retains the regulatory treatment is the one in which collateral reuse cannot be measured, and the venue that escapes that treatment is the one in which reuse can be read directly from the ledger. Supervisory reach and empirical visibility, which the precrisis literature could treat as moving together, now point in opposite directions.

TABLE 14 Asset and liability swaps from wholesale CBDC issuance (scenario 1).

(a) Central bank: wholesale CBDC issuance	
Assets	Liabilities
Treasury securities	Reserves ↓ Fiat money Wholesale CBDC ↑
(b) Commercial bank: holding wholesale CBDC	
Assets	Liabilities
Cash	Bank deposits
Reserves ↓	Stablecoins issued
Treasury securities	
Asset securities	
Loans	
Cryptocurrencies (as investment)	
Stablecoins (held as asset)	
Wholesale CBDC ↑	

3.8 Stage 6: the advent of CBDC

In the final stage, the central bank itself enters the digital currency space, introducing what are known as CBDCs. This fundamentally alters the structure of the financial system. We analyze three potential scenarios.

3.8.1 Scenario 1: wholesale CBDC only

In this scenario, the central bank issues a wholesale CBDC (wCBDC), a tokenized form of central bank money for use exclusively by financial institutions. Its purpose is to enhance the efficiency and speed of interbank payments and settlements.

A commercial bank acquires wCBDC not through an expansion of the central bank’s balance sheet but through a direct swap of its existing traditional reserves. This results in a compositional shift, not a size change, on both balance sheets (Table 14). For the bank, it is an asset swap; for the central bank, a liability swap.

3.8.2 Scenario 2: retail CBDC and the threat of disintermediation

In this scenario, the central bank issues a retail CBDC (rCBDC), giving the public direct access to its liabilities. This poses a significant disintermediation risk to

TABLE 15 Balance sheet neutrality when retail CBDC substitutes for physical cash (scenario 2, case 2a).

(a) Central bank: rCBDC replaces cash	
Assets	Liabilities
Treasury securities	Reserves Fiat money ↓ Retail CBDC ↑
(b) Household: retail CBDC substitutes cash	
Assets	Liabilities
Cash ↓ Bank deposits Treasury securities MMFs Cryptocurrency Retail CBDC ↑	Loans

commercial banks. The magnitude of this risk depends critically on whether the rCBDC substitutes for physical cash or for commercial bank deposits.

3.8.2.1 Case 2a: substitution for cash. If households merely swap physical cash for rCBDC, the impact on the banking system is minimal. The central bank’s liabilities shift from fiat money to rCBDC, and the household’s assets shift from cash to rCBDC, leaving the commercial bank’s balance sheet unaffected (Table 15).¹⁵

3.8.2.2 Case 2b: substitution for deposits. This is the high-risk scenario. If households view rCBDC as a perfect substitute for bank deposits (offering superior safety with no credit risk), they may withdraw their funds from commercial banks. As research by Bouis *et al* (2024) highlights, this would trigger a significant contraction of the banking sector’s balance sheets. For every dollar moved from a deposit to an rCBDC, the commercial bank loses a dollar of funding and a dollar of its most liquid asset: reserves.

¹⁵ In this paper we assume that commercial banks can accept and handle rCBDCs. This implies banks need to upgrade their technology and businesses quickly.

TABLE 16 Three-entity balance sheet contraction when retail CBDC substitutes for bank deposits (scenario 2, case 2b).

(a) Central bank: rCBDC replaces deposits	
Assets	Liabilities
Treasury securities	Reserves ↓ Fiat money Retail CBDC ↑
(b) Household: retail CBDC substitutes deposits	
Assets	Liabilities
Cash Bank deposits ↓ Treasury securities MMFs Cryptocurrency Retail CBDC ↑	Loans
(c) Commercial bank: contraction from retail CBDC	
Assets	Liabilities
Cash Reserves ↓ Treasury securities Asset securities Loans	Bank deposits ↓

Simultaneously, the central bank’s liability composition shifts from bank reserves to retail CBDC: its balance sheet size is unchanged, but its counterparty shifts from commercial banks to households. As shown in Table 16, this three-entity chain illustrates how rCBDC adoption can simultaneously contract the commercial banking sector while expanding the central bank’s direct relationship with the public.

The severity of this disintermediation also depends on the issuance model. A direct model, where the central bank manages retail accounts, poses the greatest threat. A two-tier model, where commercial banks act as intermediaries for rCBDC wallets (similar to China’s e-CNY system (Bai *et al* 2025)) may mitigate some operational shocks but does not eliminate the fundamental risk of deposit outflows.

TABLE 17 Liability substitution under the synthetic e-CNY 2.0 model preserving bank intermediation (scenario 2, case 2c).

(a) Commercial bank: e-CNY 2.0 (no disintermediation)	
Assets	Liabilities
Reserves	Bank deposits ↓
Loans	e-CNY deposits (insured) ↑
Securities	
(b) Household: e-CNY 2.0 adoption	
Assets	Liabilities
Cash	Loans
Bank deposits ↓	
e-CNY deposits (interest-bearing) ↑	
Treasury securities	
MMFs	
Cryptocurrency	

3.8.2.3 Case 2c: the synthetic model. China’s e-CNY 2.0 framework, introduced in early 2026, fundamentally altered the risk profile of retail CBDC. In this case, the digital currency is legally and technically integrated into the commercial banking system. Unlike the direct claim model in case 2b, these digital yuan holdings are classified as commercial bank liabilities rather than direct central bank liabilities. This classification extends regulatory safeguards to the digital currency: it is covered by deposit insurance schemes, and unlike physical cash, commercial banks are permitted to pay interest on these holdings.

This structural shift effectively neutralizes the threat of disintermediation. When a household converts traditional savings into e-CNY 2.0, the transaction is recorded as a liability substitution within the commercial bank’s balance sheet (swapping “legacy deposits” for “e-CNY deposits”) rather than a withdrawal of funds. As illustrated in Table 17, the commercial bank retains the liquidity necessary for lending, ensuring that the credit creation process remains intact.

As shown in the table, the total size of the commercial bank’s balance sheet remains constant. The bank acts not just as a wallet provider but as the actual issuer of the liability, thereby preserving its funding base and avoiding the contractionary effects seen in case 2b.

3.8.3 Scenario 3: the hybrid model – a potential mitigation

This scenario involves the simultaneous issuance of both wholesale and retail CBDCs. This hybrid model could potentially mitigate the risks of disintermediation. As argued by Auer *et al* (2022), a wCBDC could serve as a highly efficient and cheap source of intraday liquidity for banks. In a stress scenario where a bank faces rapid deposit outflows due to rCBDC adoption, it could use its wCBDC holdings (or quickly acquire more from the central bank) to meet its obligations, thus preventing fire sales of other assets and containing systemic contagion.

4 CONCLUSIONS

Table 18 summarizes the structural evolution discussed across the six stages. It reveals a fundamental shift: while traditional banking relies on fractional reserves and credit multiplier effects, the introduction of nonbank digital assets (stages 2–4) often creates leakage or narrow-banking constraints. However, the counter-response by banks (stage 5) and the integrated CBDC model (stage 6, case 2c) demonstrate how the banking sector is adapting to retain its centrality in credit intermediation.

Table 19 maps the primary propagation mechanisms triggered when a stress event originates in one asset class and transmits to other sectors of the financial system.

A number of transmission chains are worth highlighting.

- (1) *The stablecoin–MMF–Treasury nexus.* If a fiat-backed stablecoin faces mass redemptions (as during the March 2023 USDC depeg), the issuer must liquidate its MMF holdings, which forces the MMF to sell short-term Treasuries, depressing prices and tightening conditions in the repo market (a channel that connects retail crypto users to sovereign debt markets).
- (2) *The RWA–DeFi–stablecoin loop.* When tokenized Treasuries are used as collateral in DeFi protocols to mint stablecoins, a default or devaluation of the underlying RWA simultaneously impairs the DeFi protocol's solvency and the stablecoin's peg, creating a feedback loop that is overlooked in the literature, which treats RWA and stablecoin in isolation.
- (3) *The permissionless token cascade.* Bank-issued tokens on a public chain that are reused across DeFi can trigger cascading liquidations that feed back into the originating bank in the form of mass redemption requests, a digital analogue of the repo chain failures observed during the 2007–9 global financial crisis.

Several avenues for future research emerge from our framework. First, a formal quantitative model could calibrate the DeFi credit multiplier under varying collateralization ratios and stress scenarios, establishing thresholds at which on-chain leverage poses systemic risk. Second, on-chain transaction data from protocols such as

TABLE 18 Summary of the balance sheet impact of digital asset adoption on commercial banks.

Instrument	Asset side	Liability side	Net structural effect
<i>Disintermediation risks</i>			
Bitcoin (stages 2–3)	Reserves ↓	Deposits ↓	Contraction
Retail CBDC (direct)	Reserves ↓	Deposits ↓	Contraction (run risk)
RWA: financial (case 1)	Reserves ↓	Deposits ↓	Contraction (liquidity migration)
<i>Narrow banking expansion</i>			
Fiat stablecoin (stage 4)	Reserves ↑	Deposits ↑	Expansion (passive)
RWA: nonfinancial (case 2)	Minimal	Minimal	Neutral (off-balance-sheet custody)
<i>Active banking adaptation</i>			
Tokenized deposits (permissioned)	No change	Substitution	Neutral
Tokenized deposits (permissionless)	No change	Reclassification	Neutral on-balance-sheet; off-balance-sheet leverage ↑
Digital bonds	Reserves ↑	Bonds payable ↑	Expansion (active funding)
Synthetic CBDC (e-CNY 2.0)	No change	Substitution	Neutral (intermediated)

TABLE 19 Cross-asset propagation mechanisms under five stress scenarios.

Shock origin	Impacted sectors		
	Commercial banks	NBFIs/MMFs	DeFi ecosystem
Crypto crash (stages 2–3)	Reserve fluctuations from settlement flows	—	Collateral liquidation cascade
Stablecoin run (stage 4)	Deposit outflow from issuer redemptions	MMF share redemptions; Treasury fire sales	DEX liquidity drain; peg arbitrage failure
RWA default (stage 5.1)	Custodian bank reserve pressure from SPV redemptions	—	Collateral cascade in DeFi protocols (eg, MakerDAO)
Permissionless token depeg (stage 5.2)	Mass redemptions at blockchain speed; liquidity crisis	—	Cascading DeFi liquidations of bank-issued tokens
Retail CBDC adoption (stage 6)	Deposit disintermediation; funding cost ↑	Reduced demand for MMFs if rCBDC is interest-bearing	—
			Stablecoin issuers
			Crypto-backed stablecoin undercollateralization (<i>Origin</i>)
			Stablecoins minted against RWA become uncollateralized
			Contagion if bank tokens used as stablecoin collateral
			Competitive pressure on stablecoin demand

MakerDAO and Aave could be combined with traditional financial market data to empirically test the cross-asset transmission channels identified in this paper, particularly during stress episodes such as the 2023 USDC depeg. Third, our distinction between permissioned and permissionless issuance raises important questions about optimal regulatory design: specifically, how macroprudential frameworks might be extended to encompass on-chain collateral reuse without stifling the efficiency gains of tokenization.

DECLARATION OF INTEREST

The authors report no conflicts of interest. ChatGPT was used to fact check and Claude (v 4.6) was used to polish the manuscript prior to submission. However, the authors alone are responsible for the content and writing of the paper.

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REFERENCES

- Aldasoro, I., Doerr, S., Gambacorta, L., Garratt, R., and Koo Wilkens, P. (2023). *The Tokenisation Continuum*. BIS Bulletin, Volume 72. Bank for International Settlements, Basel. URL: www.bis.org/publ/bisbull72.htm.
- Armas, A., and Singh, M. (2022). Digital money and central banks' balance sheet. Working Paper WP/22/206, International Monetary Fund, Washington, DC (<https://doi.org/10.5089/9798400224072.001>).
- Auer, R., Frost, J., Gambacorta, L., Monnet, C., Rice, T., and Shin, H. S. (2022). Central bank digital currencies: motives, economic implications, and the research frontier. *Annual Review of Economics* **14**, 697–721 (<https://doi.org/10.1146/annurev-economics-051420-020324>).
- Bai, H., Cong, L. W., Luo, M., and Xie, P. (2025). Adoption of central bank digital currencies: initial evidence from China. *Journal of Corporate Finance* **91**, Paper 102735 (<https://doi.org/10.1016/j.jcorpfin.2025.102735>).
- Basel Committee on Banking Supervision (2022). Prudential treatment of cryptoasset exposures. Standards Document, December, Bank for International Settlements. URL: www.bis.org/bcbis/publ/d545.htm.
- Basel Committee on Banking Supervision (2024). Cryptoasset standard amendments. Standards Document, July, Bank for International Settlements. URL: www.bis.org/bcbis/publ/d579.htm.

- Baur, D. G., Hong, K., and Lee, A. D. (2018). Bitcoin: medium of exchange or speculative assets? *Journal of International Financial Markets, Institutions and Money* **54**, 177–189 (<https://doi.org/10.1016/j.intfin.2017.12.004>).
- BDO Advisory Services (2026). Tether International, S.A. de C.V.: assurance report according to ISAE 3000R on the financial figures and reserves report. Report, March 31, BDO Advisory Services, Milan. URL: <https://bit.ly/466KzgX>.
- Bidder, R., Jackson, T. P., and Rottner, M. (2024). CBDC and banks: disintermediating fast and slow. Discussion Paper 15/2024, Deutsche Bundesbank, Frankfurt. URL: www.bundesbank.de/en/publications/research/discussion-papers/cbdc-and-banks-disintermediating-fast-and-slow-931090.
- Bouis, R., Gelos, M. G., Nakamura, F., Miettinen, M. P. A., Nier, E., and Soderberg, G. (2024). Central bank digital currencies and financial stability: balance sheet analysis and policy choices. Working Paper WP/24/226, International Monetary Fund, Washington, DC (<https://doi.org/10.5089/9798400290794.001>).
- Brunnermeier, M. K., and Niepelt, D. (2019). On the equivalence of private and public money. *Journal of Monetary Economics* **106**, 27–41 (<https://doi.org/10.1016/j.jmoneco.2019.07.004>).
- Catalini, C., and Gans, J. S. (2020). Some simple economics of the blockchain. *Communications of the ACM* **63**(7), 80–90 (<https://doi.org/10.1145/3359552>).
- Circle Internet Group, Inc. (2026). USDC reserve report. Independent Accountants' Examination Report, March. URL: <https://bit.ly/4x5pJJY>.
- Coutts, V. (2020). Black Thursday: MakerDAO's multi collateral DAI exploitation (and the plan to recover). Blog Post, March 19, *Medium*. URL: <https://bit.ly/4x93gvB>.
- Fernández-Villaverde, J., Sanches, D., Schilling, L., and Uhlig, H. (2021). Central bank digital currency: central banking for all? *Review of Economic Dynamics* **41**, 225–242 (<https://doi.org/10.1016/j.red.2020.12.004>).
- Gorton, G. B., and Zhang, J. Y. (2023). Taming wildcat stablecoins. *University of Chicago Law Review* **90**(3), 909–971. URL: <https://lawreview.uchicago.edu/print-archive/taming-wildcat-stablecoins>.
- Gorton, G. B., and Zhang, J. Y. (2024). Bank runs during crypto winter. *Harvard Business Law Review* **14**(2), 297–338. URL: https://journals.law.harvard.edu/hblr/wp-content/uploads/sites/87/2024/10/02_HLB_14.2_Gary-B.-Gorton-Jeffery-Y.-Zhang.pdf.
- Heeb, G., and Baer, J. (2025). Big banks explore venturing into crypto world together with joint stablecoin. *Wall Street Journal*, May 25. URL: <https://on.wsj.com/4AdJxxv>.
- Kreppmeier, J., Laschinger, R., Steininger, B. I., and Dorfleitner, G. (2023). Real estate security token offerings and the secondary market: driven by crypto hype or fundamentals? *Journal of Banking and Finance* **154**, Paper 106940 (<https://doi.org/10.1016/j.jbankfin.2023.106940>).
- Kumhof, M., and Noone, C. (2021). Central bank digital currencies: design principles for financial stability. *Economic Analysis and Policy* **71**, 553–572 (<https://doi.org/10.1016/j.eap.2021.06.012>).
- Lee, M. J. (2026). Composable finance. Staff Report 1177, Federal Reserve Bank of New York (<https://doi.org/10.59576/sr.1177>).
- Liao, G. Y., and Caramichael, J. (2022). Stablecoins: growth potential and impact on banking. International Finance Discussion Paper 1334, Board of Governors of the Federal Reserve System, Washington, DC (<https://doi.org/10.17016/IFDP.2022.1334>).

- Liu, Y., and Tsyvinski, A. (2021). Risks and returns of cryptocurrency. *Review of Financial Studies* **34**(6), 2689–2727 (<https://doi.org/10.1093/rfs/hhaa113>).
- Luck, S. (2025). A historical perspective on stablecoins. Blog Post, November 1, *Liberty Street Economics* (<https://doi.org/10.59576/lse.20251001>).
- Mirdala, R. (2025). Tokenization of real-world assets: legal frameworks, market dynamics, and policy pathways for a decentralized financial future. *Journal of Applied Economic Sciences* **2**(88), 285–312 ([https://doi.org/10.57017/jaes.v20.2\(88\).09](https://doi.org/10.57017/jaes.v20.2(88).09)).
- Montgomery, J. (2025). Bitcoin pizza day: 15 years since 10,000 BTC bought two pizzas and changed everything. *Bitcoin Magazine*, May 22. URL: <https://bit.ly/46muhRq>.
- Nakamoto, S. (2008). Bitcoin: a peer-to-peer electronic cash system. White Paper, Bitcoin.org. URL: <https://bitcoin.org/bitcoin.pdf>.
- Organisation for Economic Co-operation and Development (2020). *The Tokenisation of Assets and Potential Implications for Financial Markets*. OECD Publishing, Paris (<https://doi.org/10.1787/83493d34-en>).
- Singh, M. (2011). Velocity of pledged collateral: analysis and implications. Working Paper 2011/256, International Monetary Fund, Washington, DC (<https://doi.org/10.5089/9781463923952.001>).
- Singh, M. (2026). Collateral velocity is disappearing behind a digital curtain. *Risk* **39**(1). URL: www.risk.net/7962922.
- Sockin, M., and Xiong, W. (2023). Decentralization through tokenization. *Journal of Finance* **78**(1), 247–299 (<https://doi.org/10.1111/jofi.13192>).
- US Securities and Exchange Commission (2026a). SEC clarifies the application of federal securities laws to crypto assets. Press Release 2026-30, March 17. SEC, Washington, DC. URL: www.sec.gov/newsroom/press-releases/2026-30-sec-clarifies-application-federal-securities-laws-crypto-assets.
- US Securities and Exchange Commission (2026b). Statement on tokenized securities. Statement, August 18. Division of Corporation Finance, Division of Investment Management and Division of Trading and Markets, SEC, Washington, DC. URL: www.sec.gov/newsroom/press-releases/2026-76-sec-proposes-new-regulation-crypto-assets.
- Whited, T. M., Wu, Y., and Xiao, K. (2022). Will central bank digital currency disintermediate banks? Working Paper, Social Science Research Network (<https://doi.org/10.2139/ssrn.4112644>).
- Wu, Y., Ye, W., Xu, J., and Hsu, D. F. (2025). Bitcoin price prediction using machine learning and combinatorial fusion analysis. In *2025 IEEE Conference on Artificial Intelligence*, pp. 61–68. IEEE Press, Piscataway, NJ (<https://doi.org/10.1109/cai64502.2025.00017>).
- Ye, W. (2025). A structural risk assessment framework for banks under cryptocurrencies, stablecoins and CBDC. Working Paper, Social Science Research Network (<https://doi.org/10.2139/ssrn.5311988>).
- Ye, W., and Rengifo, E. W. (2026). Balance sheet decomposition of crypto, CBDC, stablecoin and RWA. Working Paper, Social Science Research Network (<https://doi.org/10.2139/ssrn.6351498>).
- Yermack, D. (2015). Is Bitcoin a real currency? An economic appraisal. In *Handbook of Digital Currency: Bitcoin, Innovation, Financial Instruments, and Big Data*, Chuen, D. L. K. (ed), Chapter 2, pp. 29–40. Academic Press (<https://doi.org/10.1016/b978-0-12-802117-0.00002-3>).